

Art & Design

Year 5

Cambridge Primary

Student Manual



Art & Design

Year 5 · Student Manual
Prime Books

A student manual for Upper Primary, written for pupils of about nine to ten years old who are ready to draw with more control, sculpt in three dimensions, craft with fabric, and paint with purpose and feeling.

Four tables in one craft studio: drawing, sculpture, fabric and paint. Rufus the fox and Bruno the bear work beside you.

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Editorial Board

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Safety in this book: washable media, blunt tools, needles and cutting tools only under the rules your teacher sets, clay work with clean hands and clean tools. Teachers follow school policy.

Introduction

This book is a studio you can carry. It has four working tables, and each one is a unit: a drawing table, a sculpt table, a craft table and a paint table.

In earlier years you looked carefully, printed, painted, drew depth and built simple structures. Keep all of it. This year the work asks for something new: **control and choice**. Which tool, which material, which colour, which feeling, and why that one.

Every unit works the same way, so you always know where you are.

- ✓ A unit opener says what the unit is about and what you will be able to do.
- ✓ Each topic starts with a question to talk about, then a short explanation.
- ✓ Panels tell you when to watch, when to make, and when to look back at your work.
- ✓ Dashed boxes are yours. Draw, plan, stitch a sample or paint in them.
- ✓ A review at the end of each unit asks what you can do now.

At the back of the book there is a studio show to aim for, an assessment page, the answers to the closed questions, a word list, the sources used, and notes for teachers.



Picture 0.1 The Craft Studio at work. Look for the different kinds of work happening in one room: drawing at one table, clay at another, fabric and paint beyond.

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Welcome to the Craft Studio

There is a long wooden table, a jar of brushes, a lump of clay and a folded square of fabric.

This book is that room. It is called **the Craft Studio**, and it belongs to you for Year 5. Nothing in it is precious. The paper is for working on.

○ DO YOU REMEMBER?

Year 4 asked for purpose and perspective. Year 5 asks for control and choice: which tool, which material, which feeling, and why.

Four promises about this year.

- 1 You will finish things, not only start them.
 - 2 You will write down what you did, so you can do it again.
 - 3 You will look at other people's work kindly and usefully.
 - 4 You will tidy up, because a studio only works if the next person can use it.
-

Who you will meet



Rufus the fox

He draws lines until they mean something.
He measures proportions before he sculpts.
He plans a craft before the first stitch.



Bruno the bear

He mixes colour for mood as well as for matching. He presses clay slowly. He is good at saying what a piece is for and how it feels.



The craft table

Drawing tools, clay, fabric scraps, paints and sketchbooks live here. Clean as you go. Finished work dries on the side shelf.

You

Age about nine or ten. Ready to choose materials, finish projects, and talk about the process as well as the product.

How to use this book

◆ SIGNS TO LOOK FOR

- **Open the door** - a question to talk about before you begin
- **Here is how it works** - the short explanation
- **Watch me work one out** - Rufus or Bruno does one, step by step
- **Your turn** - make, draw, build or stitch, in the book
- **Did you know?** - a true fact, checked before printing
- **Go a bit further** - an extra challenge if you have time
- **Big challenge** - a longer piece of work
- **How did it go?** - a short review at the end of a unit

Dashed boxes are yours. Where a task is open making, there is no single right answer and none is printed. Where a question is closed, the answer is at the back, and you should try it before you look.



Picture 0.2 The studio signs. Look for these same marks on the pages ahead: each one tells you what kind of work comes next.

Getting set up

Your studio kit

- pencils H, HB and 2B, charcoal, a rubber and a sharpener
- a sketchbook or spare drawing paper
- air-dry clay or modelling clay, a board and blunt tools
- fabric scraps, felt, yarn, blunt needles, glue and safety scissors
- washable paints, one flat and one round brush, a palette and a water pot
- an apron or an old shirt

Room norms

- 1 Look before you touch.
- 2 Share tools, and return them to the same place.
- 3 Keep water, paint and clay contained on your board.

- 4 Needles and cutting tools follow your teacher's rules, every time.
- 5 Talk kindly about work: say what you can see, then what you wonder.
- 6 Leave the table cleaner than you found it.

▲ WARM UP YOUR HANDS

Shake your hands loosely for five seconds. Press your palms together, then open them wide like two stars. Three times. Now pick up the pencil.



Picture 0.3 The craft kit laid out on the table. Look for the water pot and the apron: the two things most often forgotten.

Working safely

Every table in this studio has one thing that needs care. None of them is dangerous if you follow the rule for it, and your teacher's rules always come first.

TABLE	THE CARE IT NEEDS
drawing	charcoal dust marks everything; wash hands before you touch fabric or paper
sculpt	blunt tools only; wipe clay from surfaces rather than sweeping it dry
craft	blunt needles, counted out and counted back in; cutting only under your teacher's rule
paint	keep water away from the edge of the table; wash brushes and stand them up to dry

Three rules for the whole room.

- 1 Tell your teacher straight away about a spill, a cut or a broken tool.
- 2 Never taste, eat or drink anything at the tables, and never blow clay dust.
- 3 Anything you take from the shelf goes back to the shelf.

■ CHECK WHAT YOU CAN DO

- 1 Where do brushes go after they are washed?
- 2 Why is clay wiped rather than swept?
- 3 What do you do first if something spills?

ANSWERS

What you will make this year

Four finished pieces, one from each table. Everything else in the book is practice towards them, and all four go into the studio show at the end of the year.

UNIT	THE FINISHED PIECE	IT IS FINISHED WHEN
1 Drawing	a personal drawing project, with its thumbnails	the subject was chosen and planned by you, and one correction is visible
2 Sculpting	a standing sculpture, and one clay work	it stands unaided for a full minute, and the joins survive being lifted
3 Fabric and craft	a mixed-media piece with a decorated edge	three materials with different jobs, stitched where it is handled
4 Painting	a purpose painting, and a mood colour study	it can be read from the distance it was made for

Write the date you finish each one on the studio journal page at the back, with the recipe or method you want to remember.

○ DO YOU REMEMBER?

Finishing is a skill on its own, and it is the one that Year 5 asks for most. A drawing that is nearly done, a pot with a weak join and a painting with one unpainted corner are all the same problem.

UNIT 1 · DRAWING TABLE

1

Exploring drawing processes

Line, tools, looking hard, and a project that is yours.

A drawing process is the way you build a drawing: marks first, tools next, observation, then a finished personal piece.

🕒 IN THIS UNIT YOU WILL

- ✓ explore mark making and line on purpose
- ✓ try different drawing tools and compare them
- ✓ strengthen observational drawing
- ✓ complete a personal drawing project

◆ ON THE TABLE TODAY

Rufus lays out an H pencil, a soft 2B, a charcoal stick and a pen.

"Same leaf," he says. "Four tools. Four different conversations with the paper."

Bruno grins. "I will start messy and finish careful. That is also a process."



Picture 1.0 The drawing table before anyone starts. Look for the four tools waiting beside one single leaf: the same subject will be drawn four ways.

UNIT 1 - TOPIC 1.1

Understanding mark making and line

□ Open the door

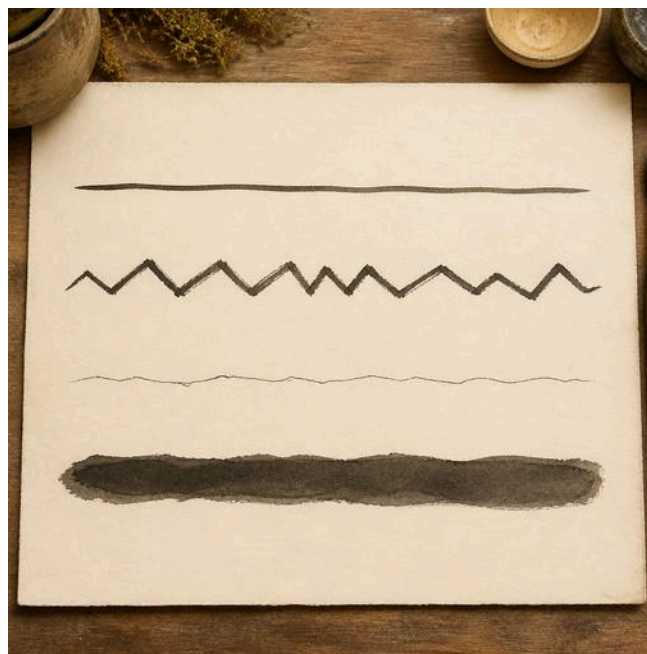
Can a line feel calm? Can a line feel angry? What would you change to make it feel that way?

► **Here is how it works**

Mark making is every trace your tool leaves on the paper: dots, dashes, hatches, smudges and continuous lines. A **line** is one mark that travels.

You already make marks without thinking. This year you choose them on purpose.

LINE IDEA	WHAT IT CAN SUGGEST
thin continuous	a careful outline, wire, a single hair
thick and bold	weight, the edge of a shadow, energy
broken or dashed	movement, a soft or unfinished edge
hatched	tone and form built without any outline
smudged	air, distance, something soft



Picture 1.1a Four line studies on one sheet. Look at how much the feeling changes when only the pressure and the speed change.

WATCH ME WORK ONE OUT

Rufus wants a jug that looks heavy.

- 1 He draws the outline first with a thin line, lightly.
- 2 He presses harder on the left edge only, where the shadow sits.
- 3 He hatches under the handle, close lines, all the same direction.
- 4 He leaves the top edge thin and pale, because light lands there.

The outline never changed shape. Only the marks did.

◆ DID YOU KNOW?

Drawing charcoal is wood that has been heated in a container with very little air, so it chars instead of burning away. That is why a charcoal stick still shows the grain of the twig it used to be.

● YOUR TURN

Fill four boxes with four different mark families. Label each one in your own words.

MARKS 1

MARKS 2

MARKS 3

MARKS 4

Which of your four would you use for the shadow under a heavy stone?



Picture 1.1b Four mark-making samples on one sheet: thin line, bold line, hatch and smudge. Look at the hatch: the tone is made only of separate lines.

► GO A BIT FURTHER

Draw one line across a whole page without lifting your pencil. Change its pressure four times on the way. Mark with an arrow where the line feels heaviest.

■ CHECK WHAT YOU CAN DO

- 1 Name two marks that build tone without an outline.
- 2 Which travels: a mark or a line?
- 3 Look at Picture 1.1a. Which of the four lines would you use for a still, empty morning?

ANSWERS

UNIT 1 - TOPIC 1.2

Experimenting with different drawing tools

□ Open the door

Which tool is bravest for a really dark shadow: an H pencil or a charcoal stick? Why do you think so?

► Here is how it works

Every tool has a strength and a warning. Choosing the tool is part of the process, so it happens before the drawing, not halfway through.

TOOL	STRENGTH	WATCH OUT
H pencil	light construction lines	can dig a groove into the paper
HB and 2B	everyday drawing, quick tone	needs sharpening often
charcoal	rich darks, soft edges	smudges if your sleeve passes over it
ink pen	one final confident line	very hard to correct
coloured pencil	layered tone, gentle mixing	slow to cover a large area

A tool can also be held in more than one way. The same charcoal stick makes a fine line on its point, a wide band on its side, and a soft field when you use it flat.



Picture 1.2a One charcoal stick held three ways, with the mark each one makes underneath it. Look for the difference between the point, the side and the flat.

● YOUR TURN

Draw the same simple object three times, each time with a different tool.

THREE TOOLS, ONE OBJECT

My three tools were:

My favourite tool for darks: _____



Picture 1.2b One cup drawn three ways: graphite, charcoal and ink. Look at the ink drawing: nothing there could be rubbed out.

◆ DID YOU KNOW?

Pencil "lead" has never been lead. It is graphite, a soft form of carbon, mixed with clay. More clay makes a harder, paler pencil, which is why an H pencil is grey and a 2B is nearly black.

| WATCH ME WORK ONE OUT

Rufus has to draw a dark doorway at the back of a bright room, and he has three tools on the table.

- 1 He rules out the H pencil: it would take twenty layers to reach that dark.
- 2 He rules out the ink pen: the edge of a shadow is soft, and ink is not.
- 3 He chooses charcoal on its side for the dark, and keeps the 2B for the edges.
- 4 He puts a scrap of paper under his hand so his sleeve cannot smudge the charcoal.

The choice took ten seconds and saved the drawing.

► GO A BIT FURTHER

Try a tool nobody planned for: a cotton bud, a folded card edge, a dry brush. Test it in a corner first. Was it useful, and what for?

My found tool:

■ CHECK WHAT YOU CAN DO

- 1 Which tool gives the darkest dark: an H pencil, a 2B pencil or charcoal?
- 2 Give one reason to choose an ink pen even though you cannot rub it out.
- 3 Why put a scrap of paper under your drawing hand when you work in charcoal?

ANSWERS

UNIT 1 - TOPIC 1.3

Developing observational drawing skills

□ Open the door

What changes if you look at the real object for ten seconds before the pencil moves at all?

► Here is how it works

Observational drawing means the real thing stays in front of you the whole time. You are not remembering it and you are not inventing it.

The loop is short and you repeat it: look, draw a little, look again, correct.

- 1 Big shapes first, lightly, and check they sit inside your paper.
- 2 Edges next, where one shape meets another.
- 3 Tone after that: which side is darker.

4 Small details last, and only the ones that matter.

You can measure without a ruler. Hold your pencil at arm's length, close one eye, and compare the height of the jug with the width of the jug. Two pencil lengths tall and one wide is a real measurement you can trust.



Picture 1.3a Measuring by eye. Look for the straight arm and the pencil held upright: the arm must not bend, or the measurement changes.

WATCH ME WORK ONE OUT

Bruno draws a pair of boots.

- 1 He measures: the pair is wider than it is tall.
- 2 He draws two soft rectangles for the whole pair, not one boot at a time.
- 3 He checks the gap between the boots. It is narrower than he first drew, so he corrects it.
- 4 Only then does he draw the laces, and he draws four, not all fourteen.

YOUR TURN

Set up three objects from the studio. Draw them where they really sit, including the space between them.

OBSERVATION DRAWING

One thing I corrected after looking again:



Picture 1.3b A still life of three objects with a careful observational drawing beside them. Look at the shapes of the gaps between the objects, not only the objects.

▲ BIG CHALLENGE

Draw a still life of three objects for twenty-five focused minutes. Nothing from memory. Turn the paper over halfway through, look at the objects for ten seconds, then turn back and correct one thing.

► GO A BIT FURTHER

Draw the same object twice from two places in the room: once from your seat, once from standing above it. Nothing about the object changed. Write down what changed in your drawing.

What changed:

■ CHECK WHAT YOU CAN DO

- 1 Put these in the order you should draw them: details, big shapes, edges, tone.
- 2 What are you comparing when you hold a pencil out at arm's length?
- 3 Why is it useful to draw the gap between two objects?

ANSWERS

UNIT 1 - TOPIC 1.4

Creating a personal drawing project

□ Open the door

What subject matters enough to you that you will finish it carefully, even on the day you would rather not?

► Here is how it works

A **personal project** is a drawing you choose, plan and finish. Four things make it a project rather than a doodle.

- 1 A chosen subject, and a reason you can say out loud.
- 2 Tool choices written down before you start.
- 3 Thumbnail sketches: three small plans, not one.
- 4 A finished drawing with a short label.

A **thumbnail** is a small quick plan about the size of a matchbox. Three thumbnails take five minutes and save you an hour.



Picture 1.4a A finished drawing beside the three thumbnails that planned it. Look at how the thumbnails try three different positions on the page.

● YOUR TURN

Plan your project here, then draw it.

Subject: _____

Why it matters to me: _____

Tools I will use: _____

THREE THUMBNAIL PLANS

FINAL DRAWING SPACE

WATCH ME WORK ONE OUT

Bruno chooses his subject.

- 1 He lists three things he could draw: his boots, the studio window, his own paw.
- 2 He picks the boots, because he can leave them on the table for a week.
- 3 Reason: "They are old and creased, and creases are interesting to draw."
- 4 Tools: 2B for the shapes, charcoal for the dark inside the boot.
- 5 He draws three thumbnails: both boots side by side, one boot close, one boot from above.
- 6 He chooses the close one, because the creases are biggest in it.

► GO A BIT FURTHER

Ask one person in your class to look at your finished drawing and tell you one thing they can see that you did not mention. Write what they said.

They noticed:

■ CHECK WHAT YOU CAN DO

- 1 How many thumbnails should you draw before a personal project?
- 2 Give one good reason for choosing a subject that stays still for a week.
- 3 What does a label on a finished drawing tell someone?

ANSWERS

Skills workshop: the drawing table

Four short drills. Ten minutes each, and they are worth repeating whenever the drawing table is set up. Do them in this order: the hand warms up, then the eye.

SKILLS WORKSHOP

Drill 1. Five pressures. Draw five short lines side by side, from the palest you can make to the darkest, in even steps.

FIVE PRESSURES

Drill 2. Hatch a ladder. Hatch four squares, each one darker than the last, using only lines. No smudging.

1 PALEST

2

3

4 DARKEST

SKILLS WORKSHOP

Drill 3. Blind contour. Look only at the object, never at your paper, and draw its outline in one slow line. It will look strange. That is the point: your eye is doing the work.

BLIND CONTOUR

Drill 4. Two-minute shapes. Draw the same object three times, two minutes each, big shapes only.

TRY 1

TRY 2

TRY 3

Which try is closest to the real thing? _____

Look at this

○ LOOK AT THIS

Turn back to **Picture 1.4a**, the drawing of the old boots with its three thumbnails.

Answer in order. There is no printed answer: these are your own readings, and the back of the book says what a good reading contains.

- 1 **What can you see?** Name three things in the drawing, without judging them.
- 2 **How was it made?** Which marks are hatching, which are single lines, which are smudges?
- 3 **What did the maker decide?** Why this arrangement, and not one of the other two thumbnails?
- 4 **What would you try?** One thing you would do differently in your own version.

MY READING

Unit 1 review

What we learned

Five words from this unit. Cover the right-hand column and say each one in your own words before you check.

WORD	WHAT IT MEANS
mark making	every trace a tool leaves: dots, dashes, hatches, smudges, lines
hatching	close parallel lines that build tone without an outline
observation	drawing from the real thing, in front of you, not from memory
thumbnail	a small quick plan, about the size of a matchbox
tone	how light or dark a part of a drawing is

What can you do now

Answer these in the book. The answers to the closed questions are at the back.

- 1 Name two marks that can make tone without any outline.
- 2 Which pencil is harder, H or 2B?
- 3 Why do artists draw big shapes before small details?

ANSWERS 1 TO 3

⊙ HOW DID IT GO?

- ☐ I can name different kinds of marks and lines.
- ☐ I compared at least three drawing tools.
- ☐ I drew from observation, correcting as I looked.
- ☐ I finished a personal drawing project.

My best process choice this unit was:

Next time I would change:

UNIT 2 · SCULPT TABLE

2

Sculpting

Form, proportion, materials, and clay that stands.

Sculpture is art in three dimensions. You build height, width and depth, and a viewer can walk around it.

🕒 IN THIS UNIT YOU WILL

- ✓ study forms and proportions
- ✓ try materials and joining techniques
- ✓ construct a 3D sculpture that stands
- ✓ make a clay work with care

◆ ON THE TABLE TODAY

Bruno rolls a clay sphere. "If the head is too big for the body," he says, "the figure tips over."

Rufus builds a card armature. "Strong inside first," he says.
"Handsome surface second."



Picture 2.0 The sculpt table with clay, card tubes and a standing figure part-built. Look at the wide base under the figure: that is what stops it falling.

UNIT 2 - TOPIC 2.1

Studying forms and proportions

□ Open the door

Is a person's head about half their height, or closer to one seventh of it?
Check on a friend before you answer.

► **Here is how it works**

A **shape** is flat. A **form** is solid: you can hold it. Almost everything you sculpt is built from five basic forms.

FORM	YOU ALREADY KNOW IT AS
sphere	an orange, a marble
cylinder	a tin of tomatoes, a tree trunk
cube	a die, a sugar lump
cone	an ice cream cone, a party hat
pyramid	a tent, a roof

Proportion is how the parts relate in size to each other. Artists often measure a figure in head heights: for an adult, about seven heads tall in total.



Picture 2.1a A clay figure beside a column of clay balls, each ball one head high. Look at the top of the stack: it reaches the top of the head, so counting the balls tells you this figure's height in head heights.

WATCH ME WORK ONE OUT

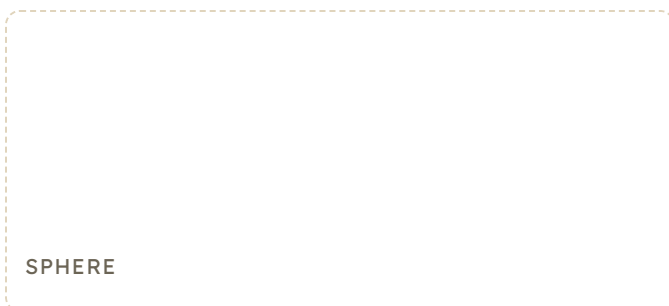
Rufus sculpts a small standing figure twenty-one centimetres tall.

- 1 He wants seven head heights, so he divides 21 by 7 and gets 3.
- 2 He rolls a head three centimetres tall and keeps it beside him as his measure.
- 3 He builds the rest of the body six head-lengths tall, which makes seven in total.
- 4 He makes the feet wider than the head, so the figure stands.

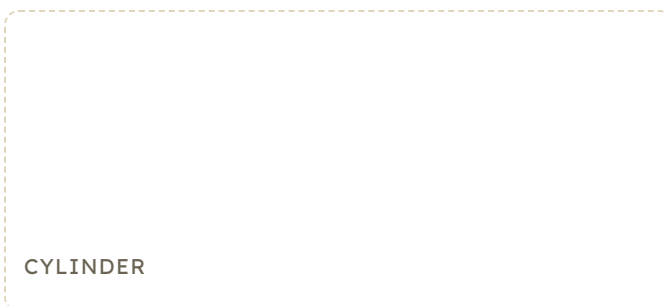
Check it: 7 heads at 3 cm each is 21 cm, the height he wanted.

YOUR TURN

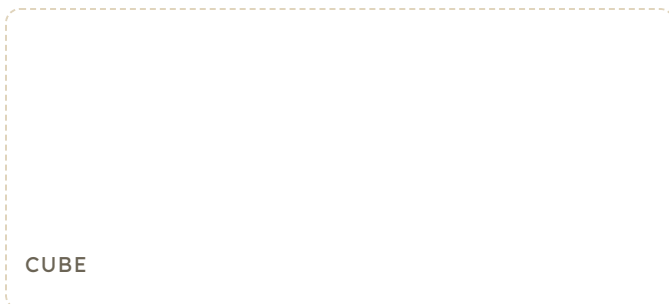
Build three basic forms from clay or card, then sketch each one here.



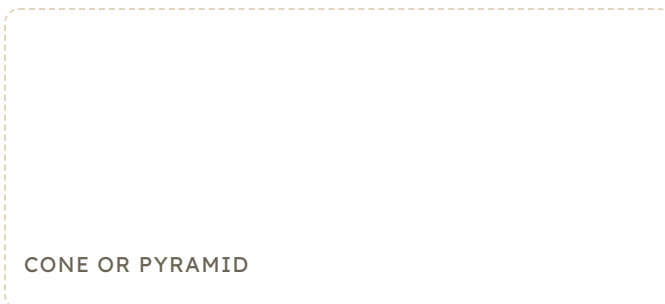
SPHERE



CYLINDER



CUBE



CONE OR PYRAMID

The hardest form to make evenly was: _____



Picture 2.1b A clay sphere, cylinder and cube on a board in soft daylight. Look at the shadow each form casts: the shadow tells you the form is round or flat.

◆ **DID YOU KNOW?**

The word terracotta comes from the Italian words for baked earth. It is clay fired in a kiln and left unglazed, which is why a terracotta plant pot still feels slightly rough and drinks up water.

■ **CHECK WHAT YOU CAN DO**

- 1 What is the difference between a shape and a form?
- 2 A figure is 24 cm tall and you want six head heights. How tall is the head?
- 3 Why do wide feet help a standing figure?

ANSWERS

UNIT 2 - TOPIC 2.2

Experimenting with materials and techniques

□ Open the door

When would card be a better choice than clay? When would clay win easily?

► Here is how it works

A sculpture is a material plus a way of joining it. Choose both before you begin.

MATERIAL	GOOD FOR	JOIN IT BY
clay	round organic forms, faces, animals	scoring and slip, or a press join
card	flat planes, buildings, boxes	slotting, a glue tab, paper tape
wire or pipe cleaner	line travelling through space	twisting the ends together
clean found objects	mixed sculpture, texture	glue, with adult help

Two joining words are worth learning properly.

- **Score and slip:** scratch both clay surfaces, wet them, press and smooth the seam.
- **Armature:** a hidden skeleton, usually wire or card, that holds the shape up.



Picture 2.2a An armature standing on its base with no clay on it yet. Look at how the wire is twisted, not just bent: a twist holds and a bend springs back.

● YOUR TURN

Join two different materials into one small study piece.

Materials:

Join method: _____

SKETCH OF MY JOIN

Did the join hold when you lifted the piece? _____



Picture 2.2b Card and clay joined in a small mixed study. Look at where the two materials meet: the join has been made flat so it holds.

WATCH ME WORK ONE OUT

Rufus joins a card tube to a clay base and it keeps falling off.

- 1 First try: he pushes the tube into the clay. It stands, then tips.
- 2 He works out why: only the rim of the tube touches, and a rim is almost nothing.
- 3 Second try: he cuts four tabs round the bottom of the tube and folds them outwards.
- 4 Now the tabs press flat into the clay, so the join has a whole area, not an edge.
- 5 It holds when he lifts the base.

A join needs area. That is the whole rule, in clay, card or paper.

◆ DID YOU KNOW?

The word sculpture comes from the Latin *sculpere*, to carve. The oldest meaning is taking material away, which is what a carver does, even though most school sculpture today is made by building material up.

► GO A BIT FURTHER

Make the same small form twice, once in clay and once in card. Which material suited it, and why?

Better in: _____

■ CHECK WHAT YOU CAN DO

- 1 What is an armature, and what is it for?
- 2 Name the two steps of a score-and-slip join.
- 3 Why does a folded tab hold better than the cut edge of a tube?

ANSWERS

UNIT 2 - TOPIC 2.3

Constructing a 3D sculpture

□ Open the door

Does your sculpture tell a viewer which side is the front? How can it do that without a label?

► Here is how it works

A sculpture that stands is built in an order. Skipping to the details is the most common mistake in the studio.

- 1 Say the purpose or idea in one sentence.
- 2 Build the base or armature, wider than you think you need.
- 3 Add the main volumes: the big forms, roughly.
- 4 Check it from all four sides, and from above.
- 5 Add details last, and only where they earn their place.
- 6 Test it: does it stand when the table is tapped gently?

A sculpture usually shows its front by facing something, by opening towards the viewer, or by being detailed on one side.

WATCH ME WORK ONE OUT

Bruno builds a bird on a post.

- 1 Purpose: a bird about to take off.
- 2 Base: a card disc, twelve centimetres across, weighted with clay underneath.
- 3 Volumes: a cylinder post, a sphere body, a cone tail.
- 4 He turns it round and finds the tail hides the body from the left, so he lifts the tail.
- 5 Details: two claws and one folded wing edge.
- 6 Tap test: it wobbles, so he widens the clay under the disc.

YOUR TURN

Purpose of my sculpture, in one sentence:

PLAN DRAWING, FROM THE FRONT AND FROM THE SIDE

Tap test: did it stand? _____



Picture 2.3 A finished standing card-and-clay sculpture on a wide base. Look at the base first: it is wider than the tallest part of the sculpture.

▲ **BIG CHALLENGE**

Build a standing sculpture at least as tall as your hand, from wrist to fingertip. It must stand on its own on a flat table for a full minute. Label it with its purpose.

■ CHECK WHAT YOU CAN DO

- 1 Put these in build order: details, base, main volumes, purpose.
- 2 Name two ways a sculpture can show a viewer which side is the front.
- 3 What is the tap test, and what does it tell you?

ANSWERS

UNIT 2 - TOPIC 2.4**Clay works****□ Open the door**

What do you think happens if you press wet clay onto bone-dry clay without scoring it first?

► Here is how it works

Clay is a material with habits. Learn them once and your work survives the drying shelf.

- 1 Knead the clay gently to push out the big air pockets.
- 2 Keep spare clay under a damp cloth so it does not skin over.
- 3 Score and wet every join, then smooth the seam closed.
- 4 Hollow out any form thicker than your thumb, so it dries evenly.
- 5 Dry slowly, away from a radiator or direct sun.
- 6 Paint only when your teacher says the piece is properly dry.

Thick solid clay cracks because the outside shrinks while the inside is still wet. Hollowing is not laziness; it is what stops the crack.



Picture 2.4a A thick clay form cut open to show the hollow inside. Look at the wall thickness: it is even all the way round, about the width of a finger.



Picture 2.4b Scoring the edge of a clay slab before joining. Look for the criss-cross marks and the dish of water: both are needed for the join to hold.

◆ DID YOU KNOW?

In Caldas da Rainha, in Portugal, the Bordallo Pinheiro factory has been making moulded faience since 1884. Its plates and dishes are shaped as cabbage leaves, fish and swallows, so the useful object and the sculpture are the same thing.

● YOUR TURN

Make a small clay pot or animal form with at least one strong join.

DRAWING OF MY CLAY WORK

My join was made by: _____

I hollowed it because:

WATCH ME WORK ONE OUT

Bruno makes a small pot with a lid that fits.

- 1 He rolls the pot first and lets it firm up for ten minutes.
- 2 He measures across the top of the pot with a strip of card, and marks the width on the strip.
- 3 He cuts the lid using the marked strip, not by eye.
- 4 He hollows the underside of the lid so it is not thicker than his thumb.
- 5 He dries both together, side by side, so they shrink by the same amount.

CHECK WHAT YOU CAN DO

- 1 Why keep spare clay under a damp cloth?
- 2 How thick is too thick for a solid clay form?
- 3 Why dry a pot and its lid side by side?

ANSWERS

Skills workshop: the sculpt table

Four drills at this table. The first three are about whether a thing holds together, and the fourth is about what a viewer walking round the table actually sees.

SKILLS WORKSHOP

Drill 1. The pinch test. Pinch a walnut of clay into a small bowl with even walls. Turn it round: are the walls the same thickness all the way?

DRAW YOUR BOWL FROM THE SIDE

Drill 2. Three joins in three minutes. Make a score-and-slip join, a press join and a card tab join. Then pull each one gently.

JOIN

HELD?

WHAT FAILED FIRST

score and slip

press

card tab

SKILLS WORKSHOP

Drill 3. Stand it up. Build the tallest thing you can that still stands on its own, from one sheet of card and no glue. Slots only.

MY TALLEST STANDING STRUCTURE

How tall did it stand, in centimetres? _____

SKILLS WORKSHOP

Drill 4. Draw it from three sides. Put any small sculpture on the table. Without moving it, draw it from the front, from the side, and from above.

FRONT

SIDE

FROM ABOVE

Which view surprised you most? _____

A sculptor checks all three before deciding a piece is finished, because a viewer walking round the table sees every one of them.

Look at this

○ LOOK AT THIS

Turn back to **Picture 2.3**, the finished standing sculpture on its wide base.

- 1 **What can you see?** Name the basic forms you can find in it.
- 2 **How was it made?** Where do two materials meet, and how were they joined?
- 3 **What did the maker decide?** Why is the base that wide, and why is the detail where it is?
- 4 **What would you try?** One change that would make it look as if it were about to move.

MY READING

Unit 2 review

What we learned

WORD	WHAT IT MEANS
form	a solid three-dimensional shape you can hold
proportion	how the parts of something relate in size to each other
armature	a hidden skeleton of wire or card that holds a sculpture up
score and slip	scratch and wet two clay surfaces so they join and hold
terracotta	fired unglazed clay: Italian for baked earth

What can you do now

Answer in the book. The closed answers are at the back.

- 1 Name the five basic forms.
- 2 About how many head heights tall is an adult figure?
- 3 Why is a thick solid clay form likely to crack as it dries?

ANSWERS 1 TO 3

⦿ HOW DID IT GO?

- ☐ I can name the basic 3D forms and find them in real objects.
- ☐ I measured proportions instead of guessing them.
- ☐ I joined materials on purpose and tested the join.
- ☐ I built a sculpture that stands on its own.
- ☐ I used clay carefully, hollowing thick forms.

The join I trust most is: _____

UNIT 3 · CRAFT TABLE

3

Exploring fabric and other crafts

Texture, pattern, stitch, mixed media and a decorated edge.

Craft is making with your hands and a material that has its own mind. Fabric bends, frays and folds, and a good crafter works with that.

🕒 IN THIS UNIT YOU WILL

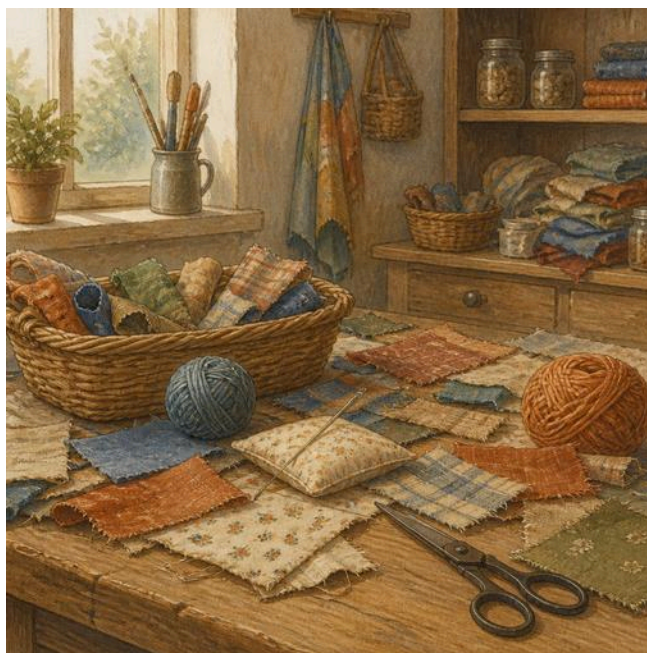
- ✓ explore fabric textures and patterns
- ✓ learn basic craft techniques, including two stitches
- ✓ make a craft project with mixed media
- ✓ explore decorative techniques on an edge

◆ ON THE TABLE TODAY

Rufus spreads out four scraps: hessian, felt, cotton and a piece of old blanket.

"Close your eyes and touch them," he says. "You will know which is which before you look."

Bruno keeps his eyes open. "I want to know which one will fray," he says, "because that changes what I make."



Picture 3.0 The craft table with fabric scraps, yarn, a blunt needle and scissors. Look at the cut edges: some are clean and some are already fraying.

UNIT 3 - TOPIC 3.1**Exploring fabric textures and patterns****□ Open the door**

Can you tell two fabrics apart with your eyes closed? What is your hand actually noticing?

► Here is how it works

Texture is how a surface feels, and how it looks as if it would feel. **Pattern** is a design that repeats.

Fabric has texture for a reason: it is built from threads crossing each other, and the size of those threads changes everything.

FABRIC	TEXTURE	PATTERN OFTEN USED
hessian	rough, open, very visible weave	none, the weave is the pattern
felt	dense, soft, no weave to see	printed or cut shapes
cotton	smooth, light, easy to fold	stripes, checks, small repeats
blanket wool	thick, warm, slightly hairy	plain or wide bands

A repeat can be made in three simple ways: side by side, offset like brickwork, or turned round each time.



Picture 3.1 Three fabric samples with different weaves. Look closely at the hessian: you can count the threads, which is why it feels rough.

● YOUR TURN

Take a rubbing of two fabrics. Lay thin paper on the fabric and rub the side of a pencil across it.

RUBBING 1

RUBBING 2

Fabric 1 was: _____

Fabric 2 was: _____

The rougher rubbing came from: _____

◆ DID YOU KNOW?

Portugal grows and harvests more cork than any other country in the world. Cork comes from the bark of the cork oak, which is stripped by hand and grows back, so the same tree can be harvested again about every nine years.

| WATCH ME WORK ONE OUT

Rufus has to choose a fabric for a small bag that will carry pencils.

- 1 He rules out hessian: the weave is so open that a pencil point would push through.
- 2 He rules out thin cotton: it folds beautifully but it will not take the weight.
- 3 He chooses felt, because it is dense, it does not fray, and a cut edge stays clean.
- 4 He keeps the hessian for the outside pocket, where the rough texture is the point.

► GO A BIT FURTHER

Find a repeating pattern somewhere in your school: a tiled floor, a brick wall, a fence, a jumper. Draw one repeat of it, then say which of the three kinds of repeat it uses.

Where I found it: _____

Kind of repeat: _____

■ CHECK WHAT YOU CAN DO

- 1 Which of these frays most easily: felt, hessian or blanket wool?
- 2 Name the three simple ways a pattern can repeat.
- 3 Why is the weave of hessian easy to see?

ANSWERS

UNIT 3 - TOPIC 3.2**Learning basic craft techniques****□ Open the door**

Why does a stitch need a knot at both ends, and what happens to your work if you forget one?

► Here is how it works

Four techniques carry most of this unit. Learn them slowly and in this order.

- 1 **Threading:** use a blunt needle, a short length of yarn, and a knot at the end.
- 2 **Running stitch:** in and out along a line, gaps the same size as the stitches.
- 3 **Back stitch:** forward one gap, then back into the end of the last stitch. It makes a solid line.

- 4 **Weaving:** over and under, over and under, and change which one you start with each row.

Needles are used only under the rules your teacher sets in your studio. Point down, needle in the pin cushion when you stop, count them in and out.



Picture 3.2a Running stitch and back stitch worked side by side. Look at the back stitch: there are no gaps, which is why it can hold an edge.



Picture 3.2b A small frame loom, half woven. Look at where the weaving stops: the over-and-under path is clear along the top edge.

WATCH ME WORK ONE OUT

Bruno sews a straight line of running stitch, eight centimetres long.

- 1 He draws a faint pencil line first, so the stitches have something to follow.
- 2 He knots the yarn and comes up from underneath at the start of the line.
- 3 He makes each stitch about half a centimetre, and each gap the same.
- 4 He counts as he goes and reaches the end with eight stitches.
- 5 He knots on the underside, so the knot does not show.

YOUR TURN

Work a short line of running stitch and a short line of back stitch on a fabric scrap. Then draw what your stitches look like.

MY RUNNING STITCH

MY BACK STITCH

Which stitch would hold a bag handle on? _____

■ CHECK WHAT YOU CAN DO

- 1 Which stitch has gaps: running stitch or back stitch?
- 2 Why knot the yarn on the underside of the fabric?
- 3 In weaving, what happens if you start every row with "over"?

ANSWERS

UNIT 3 - TOPIC 3.3

Creating a craft project with mixed media

□ Open the door

If a piece uses fabric, card, paint and yarn, what holds it all together, and what holds the idea together?

► Here is how it works

Mixed media means more than one material in one finished piece. It works when the materials have different jobs and a plan says which is which.

- Background: the largest piece, usually fabric or card.
- Shapes: felt, paper or painted card, cut cleanly.
- Line: yarn, stitch or drawn marks.
- Fixing: glue for large flat pieces, stitch where it must survive handling.

Appliqué is one very useful method: cut a shape and fix it onto a bigger cloth, then stitch round the edge so it stays.



Picture 3.3a Felt shapes laid out on the backing cloth with the pin cushion beside them. Look at the arrangement: the shapes are moved around and only pinned once the position is decided.

● YOUR TURN

Plan a small mixed-media piece about the size of this page's box, then make it.

My idea in one sentence:

Background material: _____

Shapes cut from: _____

Where I will stitch instead of glue:

PLAN FOR MY MIXED-MEDIA PIECE



Picture 3.3b A finished mixed-media craft piece. Look for the three different materials and for where stitch has replaced glue.

| WATCH ME WORK ONE OUT

Bruno makes a small panel about the Aveiro salt pans, in mixed media.

- 1 Idea in one sentence: flat white salt heaps under a wide sky.
- 2 Background: pale blue cotton, because the sky is most of the picture.
- 3 Shapes: three white felt triangles for the heaps, cut cleanly.
- 4 Line: one length of grey yarn for the wooden walkway, couched down with small stitches.
- 5 Fixing: glue behind the felt to hold it while he works, then stitched round every edge, because the panel will be handled.

► GO A BIT FURTHER

Swap one material for a surprising one: a bottle cap, a bus ticket, a piece of a paper bag. Clean and flat only. Does the piece still hold together as one idea?

■ CHECK WHAT YOU CAN DO

- 1 Name the four jobs materials do in a mixed-media piece.
- 2 When should you stitch rather than glue?
- 3 What does appliqué add that drawing on the fabric would not?

ANSWERS

UNIT 3 - TOPIC 3.4

Exploring decorative techniques

□ Open the door

Where do your eyes go first on a decorated cushion: the middle or the edge?

► Here is how it works

Decoration is the finishing that makes a made thing feel cared for. On fabric it usually happens at the edge, because the edge is where the eye lands and where fraying starts.

TECHNIQUE	WHAT IT DOES	GOOD FOR
blanket stitch	locks the edge with a row of loops	felt, blanket wool, bag tops
fringe	cuts the edge into strips on purpose	hessian, thick cotton
beading	threads beads along a hem	small panels, hangings
appliqué border	repeats one cut shape all round	banners, cushion fronts

Decoration is a choice, not an extra. Too much and the object disappears under it.



Picture 3.4a Three finished edges: blanket stitch, a cut fringe and a beaded hem. Look at which edge would survive being pulled at.

● YOUR TURN

Decorate one edge of a fabric scrap using one of the four techniques.

Technique I chose: _____

Why it suits my fabric:

MY DECORATED EDGE



Picture 3.4b A decorated craft panel finished at every edge. Look at how the decoration frames the middle instead of covering it.

◆ DID YOU KNOW?

The National Tile Museum in Lisbon is inside the Madre de Deus convent, begun in 1509 for Queen Leonor. Its tiles are laid as repeating patterns across whole walls, so the pattern is the decoration and the surface at the same time.

► GO A BIT FURTHER

Decorate the same edge twice on two scraps of the same fabric, using two different techniques. Ask someone which one they would choose for a cushion in a busy room, and write down the reason they give, not only their choice.

They chose: _____

Their reason: _____

■ CHECK WHAT YOU CAN DO

- 1 Which decorated edge would survive being pulled at hardest?
- 2 Give one reason to decorate an edge rather than the middle.
- 3 What goes wrong when a piece is over-decorated?

ANSWERS

Skills workshop: the craft table

Five drills, and the first one is the one everybody skips.

SKILLS WORKSHOP

Drill 1. Thread and knot, five times. Thread a blunt needle and knot the end, five times, against the clock. It is a skill and it gets faster.

How many seconds by the fifth try? _____

Drill 2. The stitch ladder. Work four short lines on one scrap: running stitch with big gaps, running stitch with small gaps, back stitch, and a row of separate straight stitches.

DRAW YOUR FOUR ROWS

SKILLS WORKSHOP

Drill 3. Weave a coaster. Cut a card loom the size of your palm and weave until it is full. Change the yarn twice, so the piece has three bands.

MY WOVEN COASTER

Which band is tightest, and why?

SKILLS WORKSHOP

Drill 4. Fray on purpose. Cut a square of hessian. Fray one edge to exactly one centimetre and leave the opposite edge clean. Measure it, do not guess.

MY MEASURED FRAY

Drill 5. Ten pins. Pin a felt shape onto backing cloth with ten pins, all pointing the same way, all the same distance apart. Pinning badly is why appliqué shifts while it is stitched.

MY PINNED SHAPE

Look at this

○ LOOK AT THIS

Turn back to **Picture 3.3b**, the finished mixed-media craft piece.

- 1 **What can you see?** Name every different material you can find.
- 2 **How was it made?** Which parts are stitched and which are glued?
How can you tell?
- 3 **What did the maker decide?** Which material was given the job of the line, and why that one?
- 4 **What would you try?** One material you would swap, and what it would change.

MY READING

Unit 3 review

What we learned

WORD	WHAT IT MEANS
texture	how a surface feels, or looks as if it would feel
back stitch	a stitch with no gaps, worked backwards into the last stitch
appliqué	a cut shape fixed onto a larger cloth and stitched round
mixed media	more than one material used in one finished piece
repeat	one unit of a pattern, and the way it is arranged again and again

What can you do now

Answer in the book. The closed answers are at the back.

- 1 Which stitch makes a solid line with no gaps?
- 2 What does appliqué mean?
- 3 Name one reason to decorate the edge of a fabric piece.

ANSWERS 1 TO 3

⊙ HOW DID IT GO?

- ☐ I can describe the texture of at least three fabrics.
- ☐ I worked running stitch and back stitch.
- ☐ I planned and made a mixed-media piece.
- ☐ I decorated an edge on purpose.

The technique I would use again is: _____

UNIT 4 · PAINT TABLE

4

Painting for purpose and expression

Mixing, brushwork, texture, a painting with a job, and colour that carries feeling.

A painting can do a job and carry a feeling at the same time. This unit asks you to decide both before you open the paints.

⦿ IN THIS UNIT YOU WILL

- ✓ mix colours and control your brush
- ✓ make texture and surface effects with paint
- ✓ develop a painting for a real purpose
- ✓ use colour to carry a feeling

◆ ON THE TABLE TODAY

Bruno has mixed a green he likes and used it all.

"Write down what you put in it," says Rufus. "Two parts yellow, one part blue, one brush tip of white."

Bruno writes it on the edge of his paper. "Now I can make it again on Thursday," he says.



Picture 4.0 The paint table with a palette, water pot and brushes of three sizes. Look at the palette: the mixed colours are kept apart, so none of them turn to mud.

UNIT 4 - TOPIC 4.1

Exploring colour mixing and brush techniques

□ Open the door

You have red, yellow, blue, black and white. How many different colours can you actually make?

► Here is how it works

The three **primary** paint colours are red, yellow and blue. Mixing two primaries gives a **secondary** colour.

MIX	RESULT
red and yellow	orange
yellow and blue	green
blue and red	purple

Two more words do most of the work in a real painting.

- A **tint** is a colour with white added, or with water added in watercolour. It gets lighter.
- A **shade** is a colour with a little black added. It gets darker.

Add white or water a little at a time. You can always lighten a colour again, but you cannot easily take white out.



Picture 4.1a One blue lightened step by step with water. Look along the row: each square is a tint of the square before it.

Your brush has techniques too, and they are worth practising on a scrap first.

BRUSH TECHNIQUE	HOW	USE IT FOR
wash	plenty of water, wide flat brush, one direction	skies, large calm areas
dry brush	almost no water, drag the brush	rough wood, fur, worn walls
stipple	dab the tip straight down, again and again	leaves, gravel, crowds
fine line	tip only, hand steady, slow	edges, stems, small details



Picture 4.1b Four brush marks in the same green: wash, dry brush, stipple and fine line. Look at how different one colour can look, depending only on the brush.

WATCH ME WORK ONE OUT

Rufus mixes the green of an olive leaf.

- 1 He starts with yellow, because the leaf is more yellow than blue.
- 2 He adds blue on the brush tip, mixes, and tests it on scrap paper beside the leaf.
- 3 It is too bright, so he adds the smallest touch of red to calm it.
- 4 He writes the recipe down: three parts yellow, one part blue, a touch of red.

YOUR TURN

Mix and paint the three secondary colours, then one tint and one shade of your favourite.

ORANGE

GREEN

PURPLE

A TINT AND A SHADE

My recipe for the green I mixed:



Picture 4.1c Colour mixing on a palette, with the mixes tested on scrap paper. Look at the scrap: testing before painting is part of the process.

◆ DID YOU KNOW?

At Serralves in Porto, the Museum of Contemporary Art opened on 6 June 1999 in an eighteen-hectare park. The building, by the Portuguese architect Álvaro Siza, is mostly white, so the colour a visitor sees is whatever the art brings into the room.

■ CHECK WHAT YOU CAN DO

- 1 Which two primaries make green?
- 2 You have made a colour too dark. What do you add, and how much at a time?
- 3 Which brush technique would you use for gravel: a wash or stippling?

ANSWERS

UNIT 4 - TOPIC 4.2

Experimenting with texture and effects

□ Open the door

How could you make paint feel rough, when paint is wet and smooth?

► Here is how it works

Texture in painting comes from the tool, the amount of water, and sometimes from something pressed into the paint.

- 1 **Sponge print:** dab a damp sponge in paint and press. Rough and open.
- 2 **Scratch back:** paint a layer, and while it is wet, scratch lines with a stick end.
- 3 **Salt scatter:** on a wet watercolour wash, a few grains of salt make pale spots.

- 4 **Resist:** draw with a wax crayon first, then wash paint over. The wax keeps the paint off.
- 5 **Print from an object:** a leaf, a piece of hessian, a bottle cap.

Try each one on scrap. Keep the tests: they become your own sample book.



Picture 4.2 Paint texture tests on one sheet. Look for the scratched lines and the pale salt spots: both were made while the paint was still wet.

● YOUR TURN

Test three effects here, and name each one underneath.

EFFECT 1	EFFECT 2
EFFECT 3	THE ONE I WILL USE

The effect I will use in my painting, and where:

WATCH ME WORK ONE OUT

Rufus paints the wall of an old house in Coimbra, where the plaster is falling away.

- 1 He lays a flat wash of pale ochre for the whole wall and lets it dry.
- 2 He mixes a grey and drags it with an almost dry brush, so the paper's tooth breaks it up.
- 3 While that is still damp, he scratches three lines with the end of the brush: the cracks.
- 4 He stops. The temptation is to add more, and more would turn the wall to mud.

► GO A BIT FURTHER

Paint the same small square of tree bark twice: once with a smooth wash, once with dry brush and scratching. Which one tells the truth about bark?

■ CHECK WHAT YOU CAN DO

- 1 When does salt make pale spots: on wet paint or on dry paint?
- 2 What does a wax crayon do under a paint wash?
- 3 Name one effect that must be done while the paint is still wet.

ANSWERS

UNIT 4 - TOPIC 4.3**Developing a painting for a specific purpose****□ Open the door**

Who is going to look at your painting, and where will it hang? Does that change what you paint?

► Here is how it works

A painting with a **purpose** has a job to do for somebody. A sign, a poster for the school show, a panel for the library wall, a card for a visitor.

Four questions before the first brush mark:

- 1 What is the job?

- 2 Who will see it, and from how far away?
- 3 Where will it hang, and how big must it be?
- 4 Which colours will read clearly at that distance?

A painting seen from four metres away needs fewer, bigger shapes than a painting held in the hand. That is a design decision, not a lack of detail.



Picture 4.3a A painted sign for a school garden, hanging on a fence. Look at how few shapes it uses: it has to be read while walking past.

WATCH ME WORK ONE OUT

Mariana paints a sign for the class reading corner.

- 1 Job: show where the reading corner is.
- 2 Seen by pupils from about four metres away, from the door.
- 3 It will hang above a shelf, so it is wide and short: 40 cm by 15 cm.
- 4 Colours: dark blue shapes on a pale cream ground, because that is the strongest contrast she tested.
- 5 She paints two big shapes, an open book and a leaf, and stops there.

YOUR TURN

Plan your purpose painting, then paint it on separate paper.

The job my painting does:

Who will see it, and from how far:

Size and place: _____

PLAN, DRAWN TO THE SHAPE OF THE REAL PAINTING



Picture 4.3b A finished purpose painting in the place it was made for. Look at whether it still reads clearly from the far side of the room.

■ CHECK WHAT YOU CAN DO

- 1 Name the four questions to answer before the first brush mark.
- 2 A sign will be read from five metres away. Fewer big shapes, or many small ones?
- 3 Why does the place a painting will hang change its shape?

ANSWERS

UNIT 4 - TOPIC 4.4

How colour can influence emotions

□ Open the door

Which colours would you choose to paint a very quiet morning? Which for a noisy playground?

► Here is how it works

Colours are often grouped as **warm** or **cool**.

GROUP	COLOURS	OFTEN FEELS
warm	red, orange, yellow	close, busy, bright, urgent
cool	blue, green, grey, violet	calm, distant, quiet, cold

This is not a rule that decides for you. It is a starting point, and artists break it deliberately. A cool grey can feel restful in one painting and lonely in another; what matters is what the rest of the painting is doing.

Two more tools carry feeling:

- **Contrast:** strong light against dark feels dramatic; close tones feel gentle.
- **Amount:** one small red mark in a grey painting shouts louder than a whole red page.



Picture 4.4a The same window scene painted twice, once warm and once cool. Look at the two paintings: nothing changed except the colour family.

● YOUR TURN

Paint the same simple shape twice: once to feel calm, once to feel urgent. Use colour only, not different shapes.

CALM

URGENT

The colours I used for calm: _____

Someone else said my urgent one felt:



Picture 4.4b The same studio, painted twice: cool above, warm below. Look at the whole room in each one, not only the paintings on the table: the light, the walls and the mood all change with the colour family.

WATCH ME WORK ONE OUT

Bruno paints the same harbour twice, for a display about weather.

- 1 Feeling one: a hot afternoon. Warm colours, and he lets the shapes lose their edges in the light.
- 2 Feeling two: a cold morning. The same shapes, in blue-greys, with sharper edges.
- 3 He keeps one small warm mark in the cold painting: a red buoy. It is the only warm thing, and every visitor looks at it first.
- 4 He writes both recipes on the back of each painting.

▲ BIG CHALLENGE

Paint one small painting about a place you know well in Portugal: a street in your town, the beach in winter, a kitchen at home. Decide the feeling first, write it at the top of your plan, and choose every colour to serve it.

► GO A BIT FURTHER

Take one of your finished paintings and cover half of it with a sheet of paper. Look only at the half you can see. Does that half carry the feeling on its own? If it does not, which colour is missing from it?

■ CHECK WHAT YOU CAN DO

- 1 Name three cool colours and three warm ones.
- 2 Which shouts louder in a grey painting: one small red mark, or a whole red page?
- 3 Is "warm colours feel busy" a rule or a starting point? Give a reason.

ANSWERS

Skills workshop: the paint table

Four drills. Keep them: they are your own reference sheet for the rest of the year.

SKILLS WORKSHOP

Drill 1. Water ladder. Paint six squares of one colour, adding water each time, from strongest to palest.

WATER LADDER

Drill 2. Mix and match. Choose any object on your table. Mix its colour as closely as you can, and paint a patch beside a written recipe.

MY PATCH

SECOND TRY, CLOSER

Recipe:

SKILLS WORKSHOP

Drill 3. Four marks, one colour. Wash, dry brush, stipple, fine line, all in the same green.

WASH

DRY BRUSH

STIPPLE

FINE LINE

Drill 4. Stop early. Paint a small sky in three brush loads and then put the brush down. Count them: three, and no more.

THREE LOADS ONLY

Look at this

○ LOOK AT THIS

Turn back to **Picture 4.4a**, the same window scene painted warm and painted cool.

- 1 **What can you see?** What is identical in the two paintings, and what has changed?
- 2 **How was it made?** Find a wash, and find a mark that is not a wash.
- 3 **What did the maker decide?** Which painting would you hang in a quiet reading corner, and why?
- 4 **What would you try?** One mark you would add to the cool painting to change its feeling again.

MY READING

Unit 4 review

What we learned

WORD	WHAT IT MEANS
primary colours	red, yellow and blue, the three you mix the others from
secondary colour	a colour mixed from two primaries: orange, green, purple
tint and shade	a tint has white or water added; a shade has a little black
wash	a wide, wet, even area of paint
purpose	the job a piece of work does for somebody

What can you do now

Answer in the book. The closed answers are at the back.

- 1 Which two primary colours make purple?
- 2 What is the difference between a tint and a shade?
- 3 Give one reason a painting seen from far away should use bigger shapes.

ANSWERS 1 TO 3

⊙ HOW DID IT GO?

- ☐ I mixed secondary colours and wrote a recipe down.
- ☐ I practised four brush techniques.
- ☐ I made at least three texture effects.
- ☐ I painted something for a real purpose.
- ☐ I used colour on purpose to carry a feeling.

The colour recipe I want to keep is:

Studio assessment

Assessment in this studio is evidence, not a mark out of ten. One finished piece per unit, kept safely, plus what you can say about how you made it.

UNIT	THE EVIDENCE YOU KEEP	WHAT A GOOD PIECE SHOWS
1 Drawing	your personal drawing project, with its thumbnails	marks chosen on purpose, corrections visible, finished
2 Sculpting	a standing sculpture and one clay work	proportion measured, joins that hold, it stands
3 Fabric and craft	a mixed-media piece with a decorated edge	two stitches worked, materials chosen for a job
4 Painting	a purpose painting and a mood colour study	a recipe you can repeat, colour that serves the purpose

Talking about your work

When you show a piece, say three things in this order. It takes about a minute.

- 1 What it is for, or what you were trying to find out.
- 2 One decision you made, and why you made it.
- 3 One thing you would do differently next time.

My studio record

Fill this in as you finish each unit.

UNIT	PIECE I AM KEEPING	DATE FINISHED
1		
2		
3		
4		

⊙ HOW DID IT GO?

- ☐ I have one finished piece from every unit.
- ☐ I can talk about a decision I made, not only about the result.
- ☐ I wrote down at least one recipe or method I want to keep.

The piece I am proudest of, and why:

Keeping your work safe

A piece that is lost or bent before the show cannot be assessed, and it cannot be looked at either. Four habits are enough.

- 1 Flat, not rolled. A rolled drawing keeps its curl for weeks.
- 2 Paper work goes in one named folder, drawings face to face so the graphite does not transfer.
- 3 Clay and sculpture go on the side shelf, never in a bag, and never on a windowsill in the sun.
- 4 Fabric work is folded once, along a line you choose, not crumpled into a pocket.

Write your name on the back of everything, in pencil, in the same corner every time. A visitor at the show should be able to find out who made a piece without turning it over, so the label does that job on the front.

One more habit, for whoever wants it: offer to look after the class folder for a unit. Sort the work by unit and by name and tell your teacher what is missing. Finding out what is missing in March is useful. Finding out in June is too late.

Your studio year

Art in this school happens outside the studio as well. These are the moments the year is built around, and each one uses work you have already made.

WHEN	STUDIO ACTIVITY	WHAT YOU TAKE TO IT
Autumn term	drawing walk in the school grounds	sketchbook, two pencils, a board
Autumn term	corridor display of drawing projects	your finished Unit 1 project
Spring term	clay day, with the sculpt table set up all morning	your Unit 2 plan and a wide base
Spring term	fabric banner made by the whole class	one appliqué panel each
Summer term	painting for a real school need	your purpose painting plan
Summer term	the studio show	one piece from every unit

Getting ready for a display

- 1 Choose the piece, not the newest piece.
- 2 Trim any rough edges of paper cleanly, or mount the work on a plain card.
- 3 Write a label: your first name, the title, the material, the year.
- 4 Check the work is dry and clean, then hand it in on time.

Plan your own studio year

Fill this in with your teacher at the start of the year, so you know what is coming and what you need to have finished by then.

TERM	THE DATE	WHAT I HAVE TO HAVE READY	DONE
Autumn			
Autumn			
Spring			
Spring			
Summer			
Summer			

Two weeks before any display

- 1 Choose the piece and check it against the label list on the show page.
- 2 Finish the last edge, corner or caption. Displays expose unfinished edges.
- 3 Ask one person what they can see in it, and write down anything they get wrong.
- 4 If they misread it, that is a design problem, not a viewer problem. Fix the piece.

► GO A BIT FURTHER

Offer to help hang the display. Deciding what hangs beside what is a real design job: similar sizes together, or one big piece with three small ones around it. Try two arrangements on the floor before anything goes on the wall.

Studio show

The studio show is where the year lands. One piece from each unit, labelled, hung or placed so a visitor can walk round and understand what Year 5 did.



Picture 5.1 A small studio show laid out. Look at how the drawings, the clay work, the stitched panel and the painting are grouped so each one has room.

Four pieces, one show

- **From Unit 1:** your personal drawing project, with the thumbnails beside it.
- **From Unit 2:** your standing sculpture, on a small plain base.
- **From Unit 3:** your mixed-media piece, with its decorated edge showing.
- **From Unit 4:** your purpose painting, in the place it was made for if you can.

Your label

Every piece needs a label a visitor can read from a metre away.

Inês • Year 5
Old boots on the studio table

Graphite and charcoal on cartridge paper
2026

● YOUR TURN

Plan your show label here, then copy it neatly onto card.

Title of my piece:

Materials I used:

One sentence for a visitor:

Hanging the work

A show is designed, like everything else in this book. Four decisions do most of it.

- 1 **Height.** Hang at the eye height of the people who will look, which for a Year 5 show is lower than an adult would choose.
- 2 **Grouping.** Similar sizes in a row, or one large piece with three small ones around it. Not one of everything, evenly spaced.
- 3 **Space.** Leave more space between groups than inside a group. That is what tells a visitor where one idea ends.
- 4 **Light.** Keep work off a sunny windowsill: paper fades and clay dries unevenly.



Picture 5.2 Work going up on a display board. Look at the height of the row of drawings and at the space left between the groups.

■ CHECK WHAT YOU CAN DO

- 1 Whose eye height decides how high the work hangs?
- 2 Where should there be more space: inside a group or between groups?
- 3 Give one reason not to hang work in direct sunlight.

ANSWERS

Art around you in Portugal

You do not have to go to a gallery to look at art. Four of the things this book has taught you are already outside, within a bus ride of most Portuguese towns.

Tiles on a wall

An **azulejo** is a painted ceramic tile, and whole walls of them are laid as repeating patterns: the same three kinds of repeat you met in Unit 3. Look for where a pattern turns a corner, and for where a tile has been cut to make it fit.

The National Tile Museum in Lisbon is inside the Madre de Deus convent, begun in 1509 for Queen Leonor, and its rooms show tiles from the fifteenth century to now.



Picture 5.3 A tiled wall with a sketchbook propped in front of it. Look at the pencil copy: only one repeat has been drawn, which is all you need to record a pattern.

● YOUR TURN

Find a tiled or patterned wall near where you live. Draw **one repeat** of it here, and write where it is.

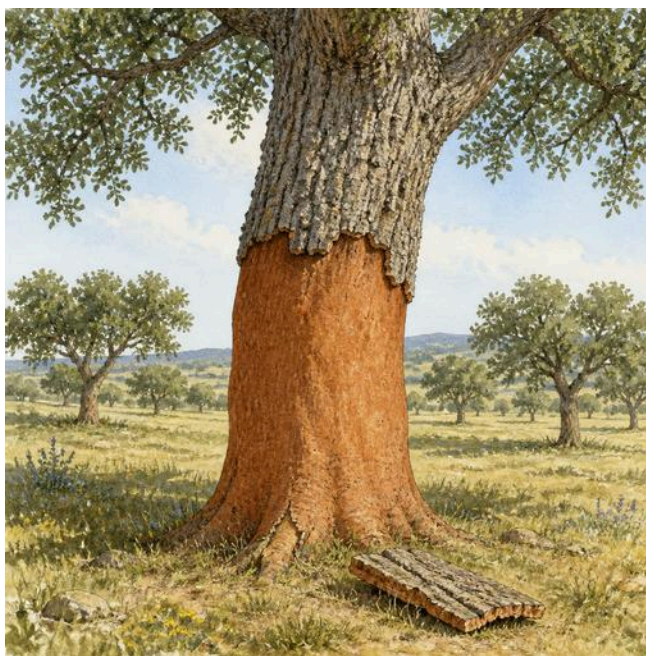
ONE REPEAT OF A PATTERN NEAR ME

Where it is:

A material that grows back

Cork comes from the bark of the cork oak. It is stripped by hand and the tree keeps growing, so the same tree can be harvested again about every nine years. Portugal harvests more cork than any other country in the world.

Cork is worth an artist's attention because it is light, it takes a pin, and it cuts cleanly, so it prints. A cork tile is a printing block waiting to happen.



Picture 5.4 A cork oak with its lower bark stripped. Look at the two colours of the trunk: smooth and reddish where the bark has gone, rough and grey where it has not.

Clay you can buy in a supermarket

In Caldas da Rainha the Bordallo Pinheiro factory has made moulded faience since 1884: plates shaped as cabbage leaves, dishes shaped as fish, jugs with swallows on them. It is the lesson of Unit 2 topic 2.4 in a shop window, where the useful object and the sculpture are the same thing.

A building that leaves the colour to the art

At Serralves in Porto, the Museum of Contemporary Art opened on 6 June 1999 inside an eighteen-hectare park. Álvaro Siza designed it almost entirely white, so the only colour in a room is whatever the art brings into it. That is a decision, exactly like the ones you make about the paper you paint on.

Three things to look for, wherever you are

You do not need any of those four places to practise looking. Every street has them.

- 1 **A repeat.** Tiles, bricks, railings, paving, a fence. Which of the three kinds of repeat is it, and where does it break?
- 2 **A made edge.** A hem, a seam, a rolled metal lip, a painted line where two colours meet. Somebody decided that edge.
- 3 **A colour that carries a job.** A red door, a green bin, a yellow line on the road. Would another colour do the same job?

► GO A BIT FURTHER

Choose one of the four places on these pages and find out one thing this book does not tell you. Write the question you asked, the answer, and where you found it. A source is part of the answer.

My question:

The answer:

Where I found it:

Answers and acceptance criteria

Try every question before you look here. Where the work is open making, there is no single right answer, so what you will find instead is what a good answer contains.

Getting set up

Working safely

- 1 Washed, standing upright to dry.
- 2 Because sweeping lifts dry clay into the air as dust.
- 3 Tell your teacher straight away.

Unit 1 - Exploring drawing processes

Topic 1.1

- 1 Hatching and smudging. Stippling with dots is also correct.
- 2 A line travels. A mark can be a single dot or dash that stays where it is.
- 3 Any of the four can be defended, but the long calm line is the usual answer, and the reason matters more than the choice.

Topic 1.2

- 1 Charcoal.
- 2 Because the line is final and confident, and nothing tempts you to fiddle with it.
- 3 So your sleeve cannot drag across the charcoal and smudge what you have drawn.

Topic 1.3

- 1 Big shapes, edges, tone, details.

- 2 The height of the thing against its width, or one part of it against another part.
- 3 The gaps are shapes too, and they are often easier to judge than the objects, so they show up a mistake early.

Topic 1.4

- 1 Three.
- 2 Because you can keep looking at the real thing across several sessions instead of remembering it.
- 3 What it is, what it was made with, and who made it.

Unit 1 review

- 1 Hatching and smudging. Both build tone without an outline. Stippling with dots is also correct.
- 2 H is the harder pencil. 2B is softer and darker.
- 3 Big shapes first fix the size and the position on the page. A mistake found then costs one line, and a mistake found after the details costs the whole drawing.

Topic 1.1, the shadow under a heavy stone

A good answer names a dark, dense mark: a heavy bold line, close hatching, or a smudge. A thin continuous line alone is not enough for weight, and you should be able to say why.

Open making in Unit 1 (four mark families, three tools, observation drawing, personal project)

A finished piece shows: marks chosen on purpose rather than by accident, at least one visible correction made after looking again, and a label saying

the subject and the tools. Thumbnails should show three different arrangements, not the same one three times.

Unit 2 - Sculpting

Topic 2.1

- 1 A shape is flat, with height and width. A form is solid, with depth as well, and you can hold it.
- 2 Four centimetres, because 24 divided by 6 is 4.
- 3 Wide feet give the figure a wide base, and a wide base does not tip.

Topic 2.2

- 1 A hidden skeleton, usually wire or card, that holds the shape up while you build on it.
- 2 Scratch both surfaces, then wet them, before pressing and smoothing the seam.
- 3 Because a folded tab gives the join an area to hold onto, and a cut edge gives it almost nothing.

Topic 2.3

- 1 Purpose, base, main volumes, details.
- 2 It can face something, open towards the viewer, or carry its detail on one side. Any two.
- 3 Tapping the table gently to see whether the sculpture stays standing. It tells you the base is wide and heavy enough.

Topic 2.4

- 1 So it does not dry out and skin over while you are working.

- 2 Thicker than your thumb.
- 3 Because they dry and shrink by the same amount, so the lid still fits afterwards.

Unit 2 review

- 1 Sphere, cylinder, cube, cone and pyramid.
- 2 About seven head heights for an adult figure.
- 3 The outside dries and shrinks while the inside is still wet, so the surface cracks. Hollowing lets it dry evenly.

Topic 2.2, did the join hold?

Either answer can be right, as long as you say what you learned. A join that failed tells you more than one that held: score both faces, wet them, press and smooth.

Open making in Unit 2 (three forms, a two-material study, a standing sculpture, a clay work)

A finished sculpture stands unaided on a flat table for a full minute, has a base wider than its tallest part, shows measured proportion rather than guessed proportion, and has joins that survive being lifted. The clay work is hollowed if it is thicker than a thumb.

Unit 3 - Exploring fabric and other crafts

Topic 3.1

- 1 Hessian, because its weave is open and its threads are thick, so they pull free easily.
- 2 Side by side, offset like brickwork, or turned round each time.

- 3 Because the threads are thick and widely spaced, so you can see and count them.

Topic 3.2

- 1 Running stitch.
- 2 So the knot is hidden underneath and the front of the work stays clean.
- 3 The rows line up on top of each other instead of alternating, so no proper weave forms and the piece falls apart when you take it off the loom.

Topic 3.3

- 1 Background, shapes, line, and fixing.
- 2 Wherever the piece will be handled, lifted or pulled.
- 3 It adds a real edge, a real thickness and a different texture, which drawing on the surface cannot do.

Topic 3.4

- 1 Blanket stitch, because it locks the edge with a row of loops.
- 2 Because the edge is where the eye lands, and it is where fraying starts. Either reason is enough.
- 3 The object disappears underneath the decoration.

Unit 3 review

- 1 Back stitch. It has no gaps, so it makes a solid line and holds.
- 2 Cutting a shape from one fabric and fixing it onto a larger cloth, then stitching round the edge.
- 3 The edge is where the eye lands, and it is where fabric starts to fray.

Topic 3.2, which stitch would hold a bag handle on?

Back stitch. A bag handle is pulled every time the bag is lifted, and running stitch has gaps that let the fabric move.

Open making in Unit 3 (rubblings, two stitch samples, a mixed-media piece, a decorated edge)

A finished craft piece uses at least three materials with different jobs, is stitched where it will be handled rather than glued everywhere, and has one decorated edge chosen to suit the fabric. Rubbings should be different enough that a reader can tell which fabric made which.

Unit 4 - Painting for purpose and expression

Topic 4.1

- 1 Yellow and blue.
- 2 Add white, or water in watercolour, a very little at a time.
- 3 Stippling.

Topic 4.2

- 1 On wet paint.
- 2 It resists the paint, so the wax lines stay pale while the wash goes round them.
- 3 Salt scatter and scratching back. Either one.

Topic 4.3

- 1 What is the job, who will see it and from how far, where will it hang and how big, and which colours read at that distance.
- 2 Fewer big shapes.

- 3 Because the space decides the height and width the work has to fit, and often whether it is wide or tall.

Topic 4.4

- 1 Cool: blue, green, grey or violet. Warm: red, orange or yellow.
- 2 The one small red mark.
- 3 A starting point. A good reason names a painting where it is not true, or explains that contrast and amount change the feeling as much as the colour family does.

Unit 4 review

- 1 Blue and red.
- 2 A tint is the colour with white or water added, so it is lighter. A shade is the colour with a little black added, so it is darker.
- 3 From far away small shapes and thin lines disappear, so fewer and bigger shapes are what a viewer can actually read.

Topic 4.2, which one tells the truth about bark?

The dry brush and scratched version, in almost every case, because bark is broken and rough and those marks are broken and rough too. A smooth wash can be right if the bark is wet, and if you can say so, that is a good answer.

Open making in Unit 4 (secondary colours, tint and shade, texture tests, purpose painting, mood study)

A finished purpose painting states its job, its viewing distance and its size before the first mark, and it can be read from that distance. A mood study changes only the colours, not the shapes, and someone else should be able

to name the feeling without being told. A colour recipe is acceptable when another pupil could mix your colour from it.

Studio show, hanging the work

- 1 The eye height of the people who will look at it, which at a Year 5 show is lower than an adult would choose.
- 2 Between groups. More space between groups than inside them is what shows where one idea ends.
- 3 Paper fades in direct sunlight, and clay dries unevenly and can crack.

The Look at this pages

There is no printed answer, because you are reading a made thing and your reading is yours. A good reading does four things, in this order.

- 1 **What can you see** names things without judging them: three materials, two colours, a shape, an edge. No "nice" and no "bad" yet.
- 2 **How was it made** points at evidence on the surface: a brush mark, a stitch, a join, a scratched line. If you cannot point at it, it is a guess.
- 3 **What did the maker decide** turns one of those into a decision with a reason: the base is wide so it will stand, the detail is on one side so the front is obvious.
- 4 **What would you try** is one specific change, not a general wish. "A warmer sky" is specific. "Make it better" is not.

The Skills workshop drills

Drills are practice, not pieces, and there is nothing to hand in. They are working properly when your fifth attempt is faster or steadier than your first, and when you can say which attempt was closest and why. Keep them: they are your own reference sheet for the rest of the year.

Studio journal

Artists keep notes, and the notes are worth more than they look. Fill a row in whenever you finish something. By the summer you will have a record of what you can do, in your own handwriting.

DATE	WHAT I MADE	MATERIALS AND METHOD	WHAT I WOULD CHANGE

Recipes worth keeping

Write down any colour mix, join or stitch you want to be able to repeat.

NAME I GAVE IT	HOW TO MAKE IT AGAIN

Studio words

WORD	WHAT IT MEANS HERE
appliqué	a cut shape fixed onto a larger cloth and stitched round
armature	a hidden skeleton of wire or card that holds a sculpture up
back stitch	a stitch with no gaps, worked backwards into the last stitch
contrast	how different two tones or colours are from each other
form	a solid three-dimensional shape you can hold
hatching	close parallel lines that build tone without an outline
mark making	every trace a tool leaves: dots, dashes, hatches, smudges, lines
mixed media	more than one material used in one finished piece
observation	drawing from the real thing, in front of you, not from memory
primary colours	red, yellow and blue, the three you mix the others from
proportion	how the parts of something relate in size to each other
purpose	the job a piece of work does for somebody
running stitch	in and out along a line, with gaps between the stitches
score and slip	scratch and wet two clay surfaces so they join and hold
secondary colour	a colour mixed from two primaries: orange, green, purple
shade	a colour with a little black added
texture	how a surface feels, or looks as if it would feel
thumbnail	a small quick plan, about the size of a matchbox
tint	a colour with white or water added
wash	a wide, wet, even area of paint

Sources and references

The scheme of work in PDF/Input sets the scope and the order of this book. Every sentence, example and illustration here is original Prime School material.

Six facts are printed in this book, each one checked before printing.

WHERE	THE FACT	CHECKED AGAINST
Unit 2, topic 2.1	terracotta is Italian for baked earth: terra, earth, and cotta, baked	Britannica, entry for terracotta; Merriam-Webster, entry for terra-cotta
Unit 2, topic 2.2	sculpture comes from the Latin sculpere, to carve	Merriam-Webster, entry for sculpture; Online Etymology Dictionary, sculpture
Unit 2, topic 2.4	the Bordallo Pinheiro factory in Caldas da Rainha has made moulded faience since 1884	Museu Bordalo Pinheiro, A National Ceramics Factory
Unit 3, topic 3.1	Portugal harvests more cork than any other country, from bark that regrows	Corticeira Amorim, Cork Oak Forest
Unit 3, topic 3.4	the National Tile Museum in Lisbon is inside the Madre de Deus convent, begun in 1509 for Queen Leonor	Museu Nacional do Azulejo, visitportugal.com
Unit 4, topic 4.1	the Museum of Contemporary Art at Serralves in Porto opened on 6 June 1999, in an eighteen-hectare park, designed by Álvaro Siza	Fundação de Serralves; Serralves museum visitor information

The charcoal and graphite notes in Unit 1 are standard material descriptions: charcoal is wood charred with very little air, and pencil cores are graphite mixed with clay, where more clay makes a harder, paler pencil.

For teachers

Mapping

The four units follow the Year 5 Art & Design scheme of work exactly, in its order. All sixteen topics are taught: 1.1 to 1.4, 2.1 to 2.4, 3.1 to 3.4 and 4.1 to 4.4. The scheme's two remaining rows, assessment and school activities, are the Studio assessment and Your studio year pages in this back matter.

Time

Each topic is planned as one lesson of teaching plus one working session. A topic occupies about four pages, so a unit is roughly six weeks at one lesson a week, and the four units fill three terms with the studio show at the end.



Picture 5.5 The end of a studio day. Look at how the brushes are washed and standing upright and the finished work is stacked flat.

A lesson sequence that works

Each topic is written as one teaching lesson plus one working session. This is the sequence the book is built for; the panels are the lesson plan.

STAGE	IN THE BOOK	ABOUT
1	Open the door	5 minutes, whole class, no materials out
2	Here is how it works	10 minutes, with the table's tools in your hands, not the pupils'
3	Watch me work one out	5 to 10 minutes, done slowly, mistakes included
4	Your turn	the rest of the lesson, and often the whole of the next session
5	Check what you can do	5 minutes, in the book, answers at the back
6	Go a bit further, Big challenge	for whoever finishes, never as homework for everybody
7	Skills workshop	a whole session per unit, repeatable whenever the table is out
8	Look at this	15 minutes, spoken, in pairs before writing

Two things are worth protecting when time is short: the **Watch me work one out** demonstration, because it is where the technique actually transfers, and the tidying at the end, because the next class inherits the room.

What to have ready

UNIT	PREPARE
1	H, HB and 2B pencils, charcoal, one still-life group per table
2	air-dry clay, boards, blunt tools, card tubes, wire or pipe cleaners, water dishes
3	fabric scraps of clearly different textures, felt, yarn, blunt needles, pin cushions
4	washable paints in the three primaries plus black and white, two brush sizes, sponges

Assessment

Evidence, not marks: one kept piece per unit and a one-minute spoken account of it. The Studio assessment page carries the criteria in pupil language, and the Answers section gives acceptance criteria for every open task, so two teachers can agree on what a finished piece looks like.

Safety

Needles, cutting tools and clay dust follow school policy. Blunt needles for all stitching, counted out and counted back in. Clay tools are blunt; wipe surfaces rather than sweeping dry clay. Any outdoor painting needs a plan agreed in advance.

Differentiation

- Shorter route: fewer mark families, a modelled thumbnail, one stitch instead of two.
- Longer route: the Go a bit further and Big challenge panels, then a second purpose painting for a real request from another class.
- Pupils who dislike open making are often unlocked by the purpose questions in topic 4.3: a job for somebody else is easier to start than a free choice.

